SCHNET: A DEEP LEARNING TOOLBOX FOR PREDICTING MOLECULAR PROPERTIES

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ABSTRACT

Accurate quantum mechanical calculations provide reliable predictions of molecular properties but are computationally prohibitive for large scale molecular dynamics simulations and the exploration of chemical space. In the work, we explore the use of atomistic neural networks, specifically SchNet, to approximate key quantum properties such as energies and dipole moments with significantly reduced computational cost. We further extend this framework toward the prediction of spectroscopic features by modeling transition dipole moments and their associated oscillator strengths. To enable this, we set up a dataset of molecular geometries paired with excitation energies and oscillator strength labels from reliable sources and develop input representations suitable for learning these quantities. The results demonstrate that learned representations can capture the underlying structure-property relationships, offering a scalable alternative to traditional quantum methods and providing a pathway toward efficient prediction of molecular spectra.
https://github.com/rarhin711/schnet_1

1 INTRODUCTION

Quantum mechanics provides a fundamental framework for describing molecular systems through the Schrödinger equation, which governs properties such as energies, geometries, dipole moments, and spectroscopic behavior. However, exact solutions are computationally intractable for all but the smallest systems due to unfavorable scaling with system size. Approximate electronic structure methods, such as density functional theory (DFT), alleviate this challenge but still exhibit significant computational cost, typically scaling as $\mathcal{O}(N^3)$ Ramakrishnan et al. (2014).

Machine learning has emerged as an efficient alternative by learning the mapping between molecular structure and properties from reference calculations Gilmer et al. (2017). Once trained, these models enable rapid predictions with significantly reduced computational expense. Among these approaches, atomistic neural networks (ANNs) are particularly effective, as they operate directly on atomic configurations Pronobis & Tkatchenko (2018). These models can be broadly categorized into descriptor-based methods and end-to-end architectures that learn representations from atomic numbers and positions Unke & Meuwly (2021), Rupp et al. (2012).

SchNet is an end-to-end atomistic neural network that models interatomic interactions using continuous-filter convolutions. Implemented within the SchNetPack framework based on PyTorch, it enables efficient and scalable learning of molecular properties Schütt et al. (2019).

In this work, SchNet is first evaluated on benchmark datasets for predicting ground-state properties such as internal energies (U) and dipole moments. The model is then extended to a new dataset (QeMFi) containing excited-state properties Westermayr & Marquetand (2021), including excitation energies and oscillator strengths, to assess its ability to generalize beyond ground-state predictions Schütt et al. (2018).

2 MODEL

2.1 ATOM EMBEDDINGS AND INTERACTION BLOCKS

The SchNet model consists of two main components: a representation block and a prediction block. The representation block maps an atomistic system, defined by atomic numbers $Z = (Z_1, \ldots, Z_n)$ and positions $R = (\mathbf{r}_1, \ldots, \mathbf{r}_n)$, to feature vectors that encode each atom's chemical environment.

Each atom i is initially represented by an embedding vector:

$$x_i^{(0)} \in \mathbb{R}^F$$

where F is the feature dimension. These embeddings are learned during training. At layer l, atomic features are denoted as $X^{(l)} = (x_1^{(l)}, \ldots, x_n^{(l)})$.

Interatomic interactions are incorporated through continuous-filter convolutions. Pairwise distances are defined as $r_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$, and feature updates are computed as:

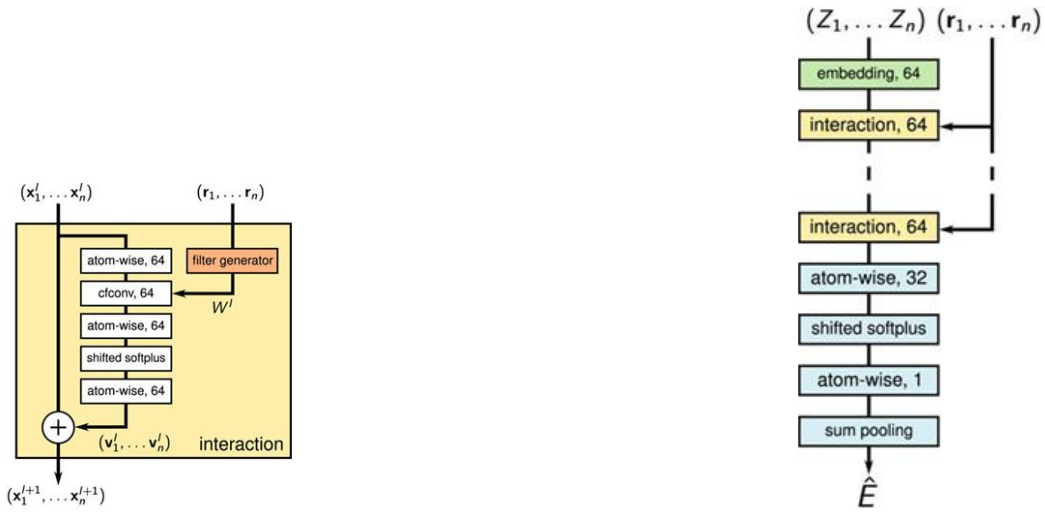
$$x_i^{(l+1)} = \sum_{j \in \mathrm{nbh}(i)} x_j^{(l)} \odot W^{(l)}(r_{ij})$$

Here, $W^{(l)}$ is a neural network that generates distance-dependent filters, enabling the model to capture spatial interactions without requiring grid-based inputs. This formulation allows SchNet to operate directly on arbitrary molecular geometries.

Following the interaction step, atom-wise fully connected layers are applied:

$$x_i^{(l+1)} = W^{(l)} x_i^{(l)} + b^{(l)}$$

which refine features independently for each atom. These layers share parameters across all atoms, ensuring that the model size does not depend on the number of atoms. The overall architecture resembles residual networks, where successive interaction blocks iteratively refine atomic representations.



(a) Interaction block

(b) Prediction block

Figure 1: Interaction and prediction block charts

2.2 PREDICTION BLOCKS

The learned atomic representations are mapped to molecular properties using prediction blocks. A common formulation expresses a property P as a sum over atomic contributions:

$$P = \sum_{i=1}^{n} p(x_i)$$

which is suitable for extensive properties such as total energy.

For intensive or vector properties, modified aggregation schemes are used. For example, the dipole moment is modeled as:

$$\boldsymbol{\mu} = \sum_{i=1}^{n} q(x_i)\,(\mathbf{r}_i - \mathbf{r}_0)$$

where $q(x_i)$ represents learned atomic charges and $\mathbf{r}_0$ is the center of mass. This formulation incorporates geometric information directly into the prediction.

2.3 Datasets

In this work, two datasets are considered. The QM9 dataset serves as a benchmark and contains approximately 133,885 small organic molecules with up to nine heavy atoms (C, O, N, F), along with ground-state properties such as internal energies and dipole moments. Data is stored and accessed through SchNetPack using the Atomic Simulation Environment (ASE), with energies and distances expressed in eV and Å, respectively.

To evaluate model generalization, a custom QeMFi dataset is introduced, containing excited-state properties, including excitation energies and oscillator strengths. Unlike QM9, where targets are scalar molecular properties, QeMFi provides multiple excited states per molecule, resulting in vector-valued targets. This shift introduces additional complexity, as the model must learn multiple correlated outputs from the same atomic representation.

3 Implementation and Methodology

3.1 Overview

This work consists of two components: (i) a benchmark study using the QM9 dataset to evaluate SchNet on ground-state properties, and (ii) a generalization study using the QeMFi dataset, which introduces excited-state targets including excitation energies and oscillator strengths. The latter serves to assess the transferability of the learned representations to a fundamentally different physical regime.

3.2 Molecular Representation and Model Architecture

Each molecule is represented as a set of atoms defined by atomic numbers Z_i and Cartesian coordinates $\mathbf{R}_i \in \mathbb{R}^3$. Atomic numbers are mapped to continuous feature vectors:

$$\mathbf{x}_i^{(0)} \in \mathbb{R}^F$$

forming the initial representation:

$$\mathcal{M} = \{(Z_i, \mathbf{R}_i)\}_{i=1}^{N}$$

Interatomic interactions are modeled using SchNet's interaction blocks. For each atom i, neighbors within a cutoff radius r_{cut} are considered:

$$\mathcal{N}(i) = \{j \mid \|\mathbf{R}_i - \mathbf{R}_j\| < r_{cut}\}$$

Pairwise distances are computed as $r_{ij} = \|\mathbf{R}_i - \mathbf{R}_j\|$ and expanded using radial basis functions:

$$\phi(r_{ij}) = [\exp(-\gamma(r_{ij} - \mu_k)^2)]_{k=1}^{K}$$

A filter-generating network produces continuous filters:

$$W^{(l)}(r_{ij}) = \text{MLP}(\phi(r_{ij}))$$

The interaction at layer l is given by:

$$\mathbf{m}_i^{(l)} = \sum_{j \in \mathcal{N}(i)} \mathbf{x}_j^{(l)} \odot W^{(l)}(r_{ij})$$

$$\mathbf{x}_i^{(l+1)} = \mathbf{x}_i^{(l)} + \mathbf{m}_i^{(l)}$$

Final molecular properties are obtained via atom-wise aggregation:

$$y = \sum_{i=1}^{N} f(\mathbf{x}_i^{(L)})$$

3.3 QM9 Dataset and Training Pipeline

The QM9 dataset contains approximately 133,885 small organic molecules with ground-state properties such as internal energy U_0 and dipole moment μ. The dataset is split into training, validation, and test sets to enable model evaluation:

$$\mathcal{D} = \mathcal{D}_{train} \cup \mathcal{D}_{val} \cup \mathcal{D}_{test}$$

Data loading is performed using an atomistic pipeline that constructs neighbor lists based on a cutoff radius. Preprocessing includes normalization of target properties, removal of mean offsets for energies, and conversion to 32-bit floating point format to ensure numerical stability. Molecules of varying sizes are handled through dynamic batching with padding and masking, enabling efficient GPU computation.

During training, batches are passed through the model, where atomic embeddings are iteratively updated via interaction blocks. The model parameters are optimized using the Adam optimizer with gradient-based updates:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

The loss function combines contributions from energy and dipole moment predictions:

$$\mathcal{L} = \lambda_1 \frac{1}{N} \sum (U_0^{pred} - U_0^{true})^2 + \lambda_2 \frac{1}{N} \sum \|\mu^{pred} - \mu^{true}\|^2$$

3.4 QeMFi Dataset and Model Adaptation

In contrast to QM9, the QeMFi dataset provides excited-state properties, including excitation energies E_n and oscillator strengths f_n. Each molecule is associated with multiple excited states, resulting in vector-valued targets:

$$\mathbf{y} = [E_1, \ldots, E_k, f_1, \ldots, f_k]$$

This introduces a data representation challenge, as the model must learn multiple correlated outputs per molecular structure. To address this, the SchNet architecture is adapted by modifying the output layer to predict multi-dimensional targets:

$$\mathbf{y}^{pred} = g\left(\sum_{i=1}^{N} \mathbf{x}_i^{(L)}\right)$$

The loss function is extended accordingly:

$$\mathcal{L} = \frac{1}{N} \sum \left[\sum_n (E_n^{pred} - E_n^{true})^2 + \sum_n (f_n^{pred} - f_n^{true})^2 \right]$$

The training pipeline remains unchanged, including batching, forward propagation, and backpropagation, ensuring consistency in model evaluation across datasets.

3.5 Evaluation Metrics

Model performance is assessed using mean absolute error (MAE) and root mean squared error (RMSE):

$$\text{MAE} = \frac{1}{N} \sum |y^{pred} - y^{true}|$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum (y^{pred} - y^{true})^2}$$

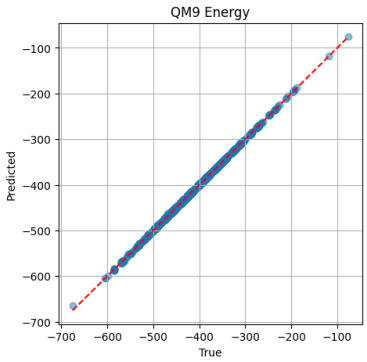
3.6 Software and Computational Framework

All implementations were carried out in Python using the PyTorch deep learning framework, which enables efficient tensor computations, automatic differentiation, and GPU acceleration. The SchNet architecture was implemented using the SchNetPack library, which provides optimized modules for atomistic modeling, including neighbor list construction, radial basis expansions, and interaction blocks Data preprocessing and dataset management were handled through SchNetPack's data utilities, while auxiliary numerical operations were performed using NumPy. Visualization of results was conducted using Matplotlib. Training was performed using the Adam optimizer, leveraging its adaptive learning rate for efficient convergence in high-dimensional parameter spaces. This integrated framework enables scalable and reproducible training of atomistic neural networks for both ground-state and excited-state property prediction.

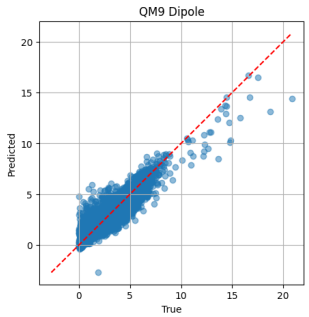
3.7 Results and Discussion

3.8 Internal energy and dipole moment prediction on QM9 dataset

The predictive performance of the model was assessed using MAE, RMSE, and parity plots for both internal energy (U_0) and dipole moment (μ). For U_0, the model shows high accuracy, with an MAE of 0.28 eV and an RMSE of 0.39 eV. The prediction improved with increasing the the number of epochs, usually from 20 to 50. The parity plot confirms this strong performance, as the predictions closely follow the ideal diagonal with minimal scatter. In contrast, dipole moment predictions are less accurate, with an MAE of 0.43 Debye and an RMSE of 0.61 Debye. The broader spread in the parity plot and the larger RMSE relative to the MAE indicate the presence of larger deviations, particularly at higher dipole values. Overall, the model predicts internal energy substantially better than dipole moment, which is expected since U_0 is a scalar property, whereas μ is more sensitive to directional charge distribution and molecular geometry.



(a) QM9 Energy parity plot



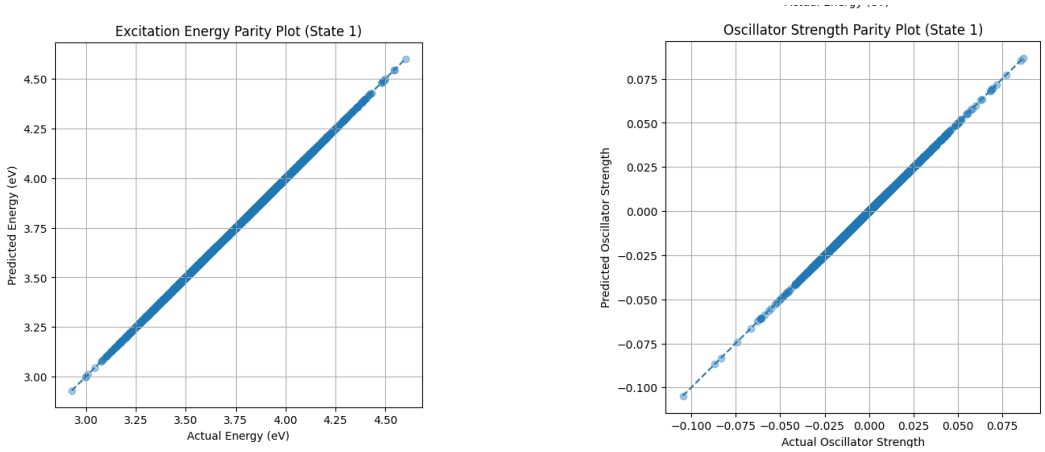
(b) QM9 Dipole parity plot

Figure 2: Parity plots for (a) internal energy and (b) dipole moment predictions.

3.9 Excitation energy, Oscillator strength prediction and spectra prediction

4 Model Performance for State 1

The model demonstrates excellent agreement with reference data for both excitation energies and oscillator strengths. As shown in the parity plots, predictions lie very close to the $y = x$ line, indicating high accuracy with negligible systematic error across the full range of values. Overall, these results show that the model achieves near-quantitative accuracy for State 1 and reliably reproduces both electronic transitions and their spectral signatures.



(a) QM9 Energy parity plot

(b) QM9 Dipole parity plot

Figure 3: Parity plots for (a) internal energy and (b) dipole moment predictions.

5 Conclusion

In this work, SchNet was successfully implemented for molecular property prediction, demonstrating strong performance across both ground- and excited-state properties. On the QM9 dataset, the model achieved high accuracy for internal energy predictions, with low MAE and RMSE values and near-ideal parity behavior. Dipole moment predictions, while reasonably accurate, showed larger deviations, reflecting the increased complexity of learning directionally sensitive properties. For excited-state properties, the model exhibited near-quantitative agreement with reference excitation energies and oscillator strengths. The close alignment in parity plots and the accurate reconstruction of spectral features indicate that the model preserves both energetic and intensity-related information consistently. Overall, these results highlight the capability of SchNet to learn physically meaningful representations of molecular systems. The model performs particularly well for scalar properties and shows strong potential for predicting spectroscopic observables, making it a valuable tool for data-driven molecular modeling and high-throughput screening applications.

LLM Acknowledgment

This work utilized large language models (LLMs), including ChatGPT (OpenAI) and Gemini (Google), to assist with some code aspects, debugging, and refinement of the computational workflow within Google Colab. These tools were also used to improve the clarity and organization of written sections. All scientific interpretations, results, and conclusions were critically evaluated and validated.

This project used an LLM for writing assistance, language editing, and code debugging support. All technical content, analysis, implementation decisions, and final verification were performed by the author.

Limitations

Despite the strong performance observed, several limitations should be noted. First, the model was trained and evaluated on relatively small molecules (QM9), which may limit its transferability to larger and more complex systems such as biomolecules or condensed-phase environments. Second, while scalar properties like internal energy are predicted with high accuracy, vector properties such as dipole moments show larger deviations, reflecting the increased sensitivity to molecular geometry and charge distribution. Additionally, the excited-state predictions were limited to a specific state and dataset, and broader validation across multiple states and chemical spaces is required. The model also does not explicitly account for environmental effects (e.g., solvent or polarization), which are important in realistic spectroscopic applications. Future improvements could include training on more

diverse datasets, incorporating environmental effects (e.g., QM/MM or EFP-based embeddings), and extending the model to predict higher excited states and more complex spectroscopic observables.

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